BRAND SENTINEL

THE DIGITAL GUARD FOR BRAND REPUTATION

A Presentation By Group 7

The Team

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The Challenge:

Managing Online Brand Perception

Companies like Apple and Google get lots of comments every second.

Negative sentiments can harm brand image very quickly.

We need a system to detect, identify and flag down negative comments early.

That's where Brand Sentinel pops in since human monitoring ain't scalable.

Objective

Turning comments into Insights

Build an NLP model to classify comments as positive, negative or neutral.

Real-Time Crisis Detection for Apple & Google @ SXSW.

Translate social chatter into actionable business intelligence

Key Questions

What is the overall sentiment towards Apple and Google?

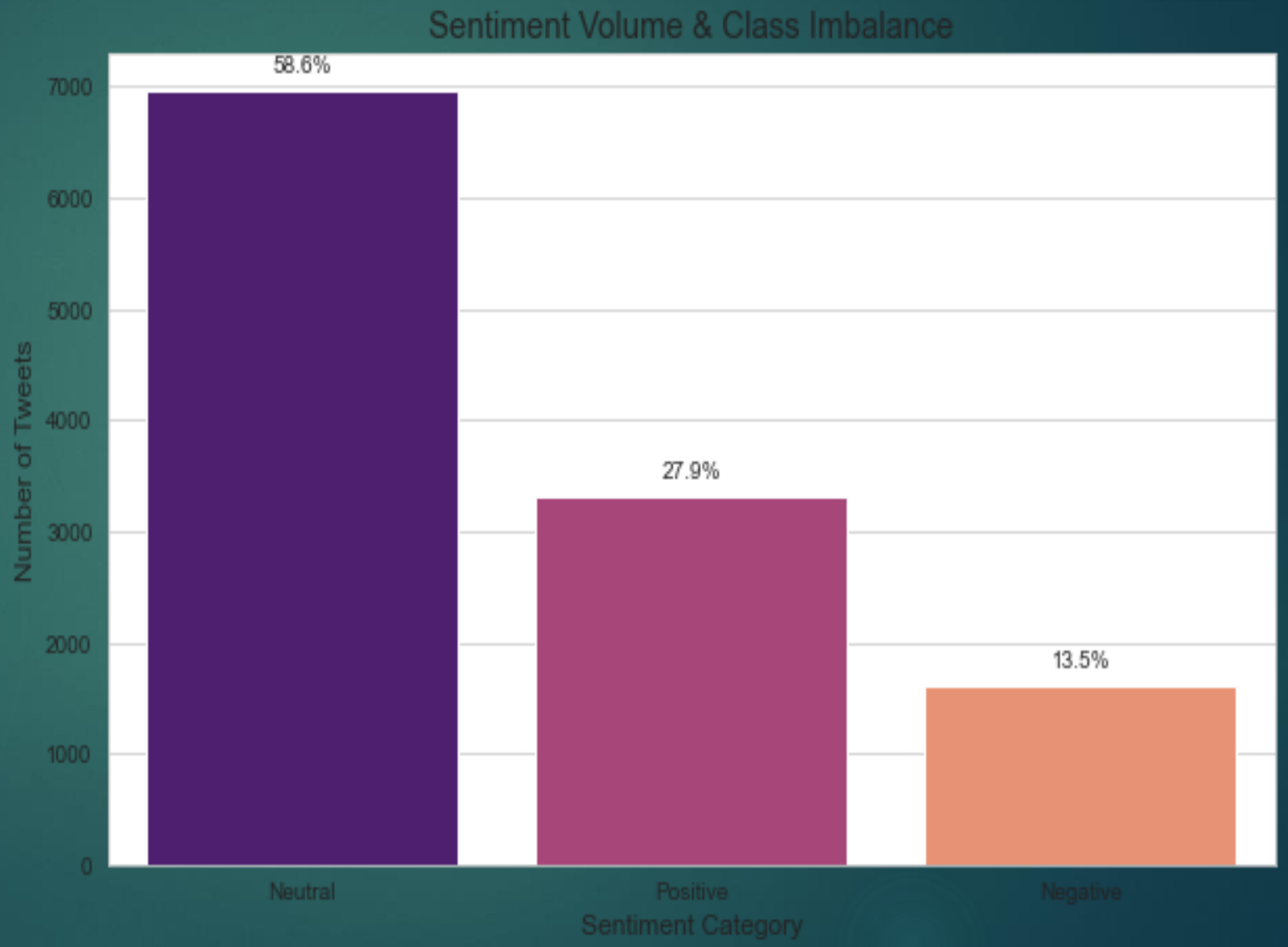
What topics trigger positive or negative reactions?

How can these insights support better decisions?

Sentiment Volume & Class Imbalance

Our dataset is heavily skewed with Neutral mentions at (58.6%) dominating the conversation.

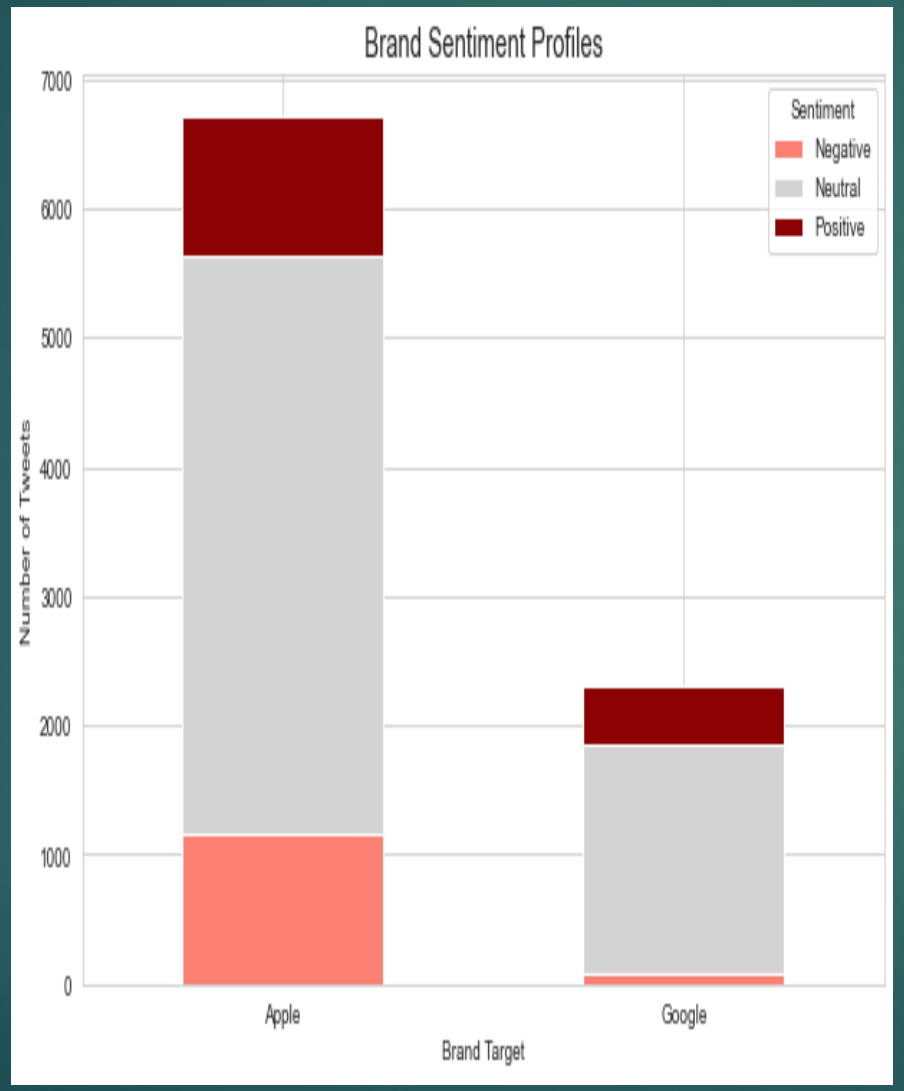
Negative sentiment of (13.5%) is the smallest class but the most critical to detect.

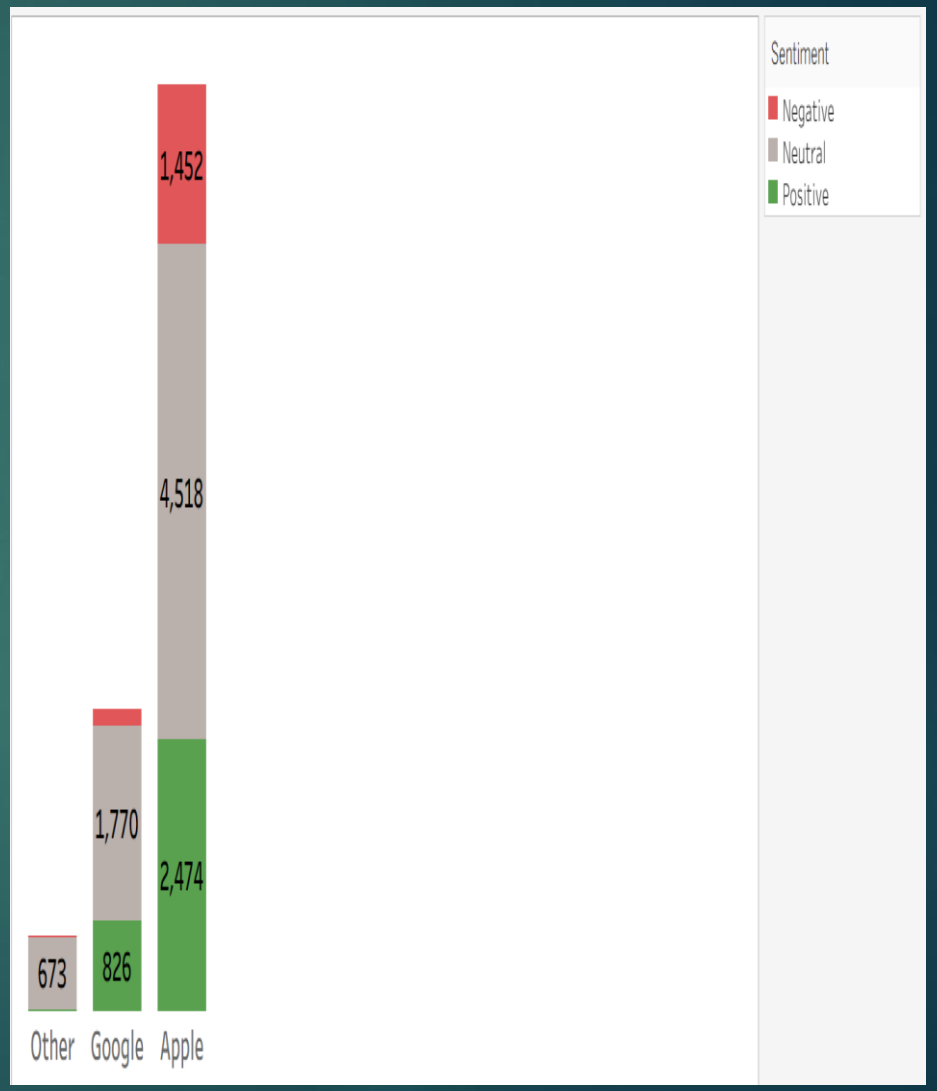


Brand Sentiment Profiles

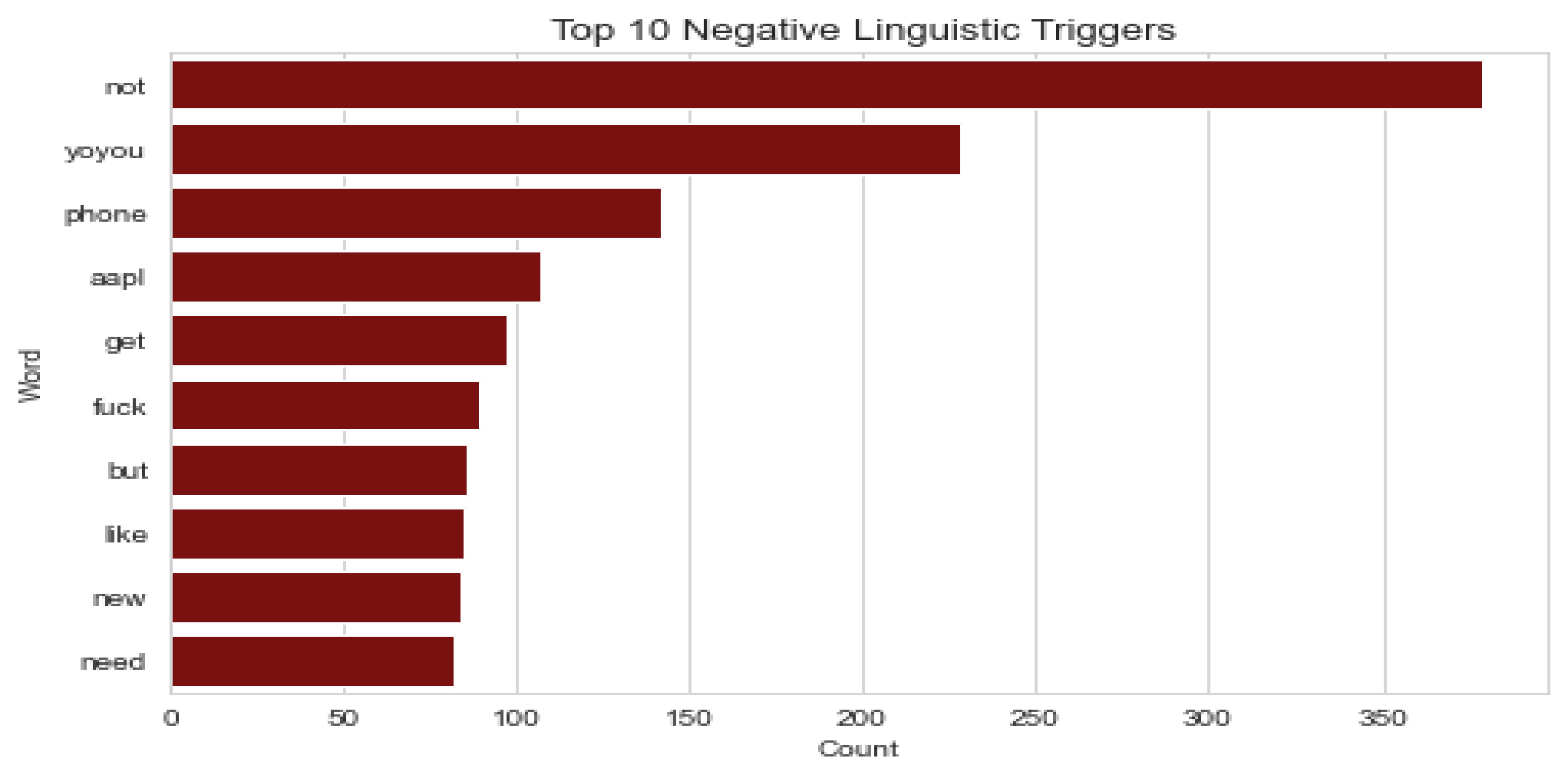
Apple's sentiment skews are more neutral while Google's sentiments are slightly mixed though with more neutrality.

The distributions helps us pinpoint areas where each company can strengthen



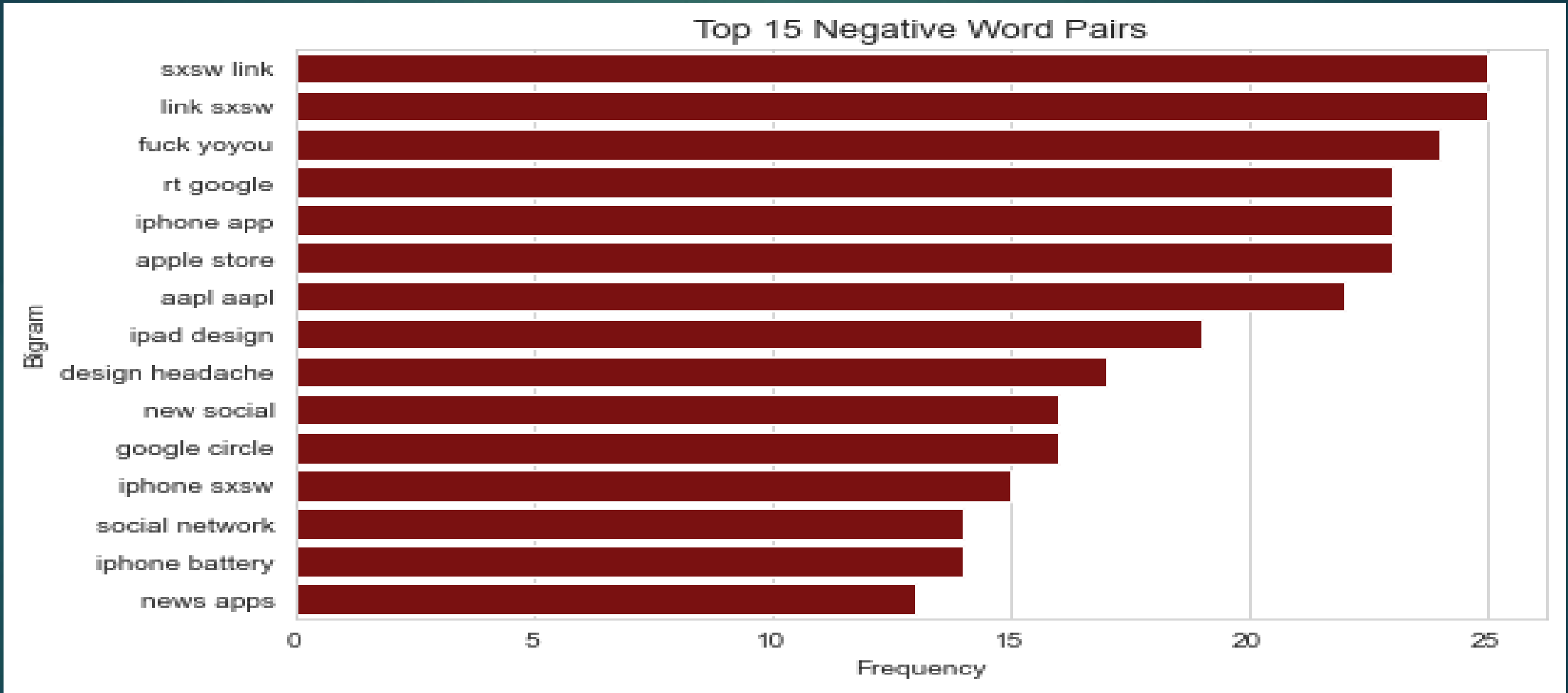


Top 10 Negative Linguistic Triggers



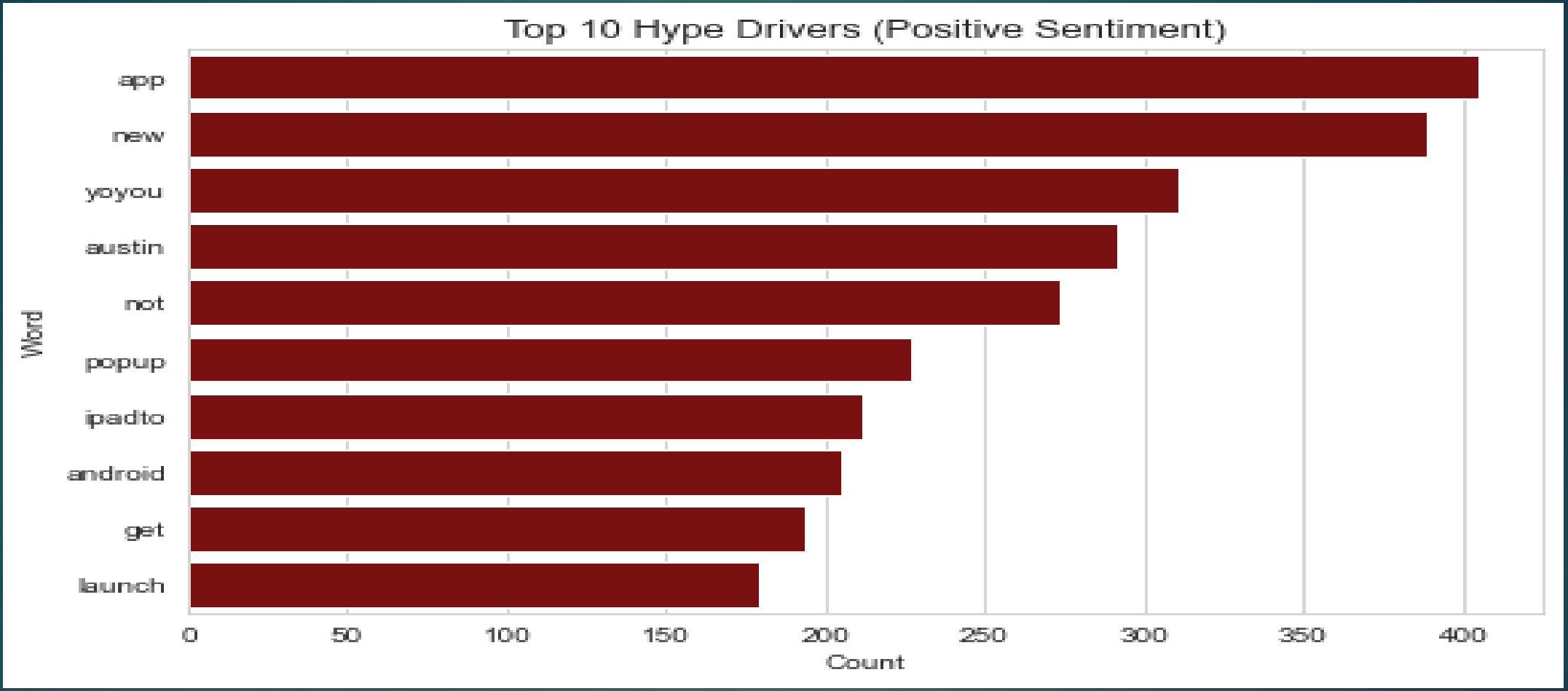
Words such as ***not*** and ***yoyou*** prove to trigger negativite comments as they have the highest counts exceeding 200 with most of the other words falling below 100 counts

Top 15 Negative Word Pairs (Crisis Phrases)



sxsw link and ***link sxsw*** both have the highest frequency for negative word pairs of 25 with ***news apps*** having the least frequency of about 13

Top 10 Hype Drivers (Positive Sentiment)



For the positive sentiment, the words ***app*** and ***new*** have the highest counts of close to 400 and above whereas the words ***get*** and ***launch*** show the least counts below 200

Model Comparison

In the final stage of Brand Sentinel, we pass real social media comments through all three models to see how they perform together.

Model Roles

Naive Bayes;

Quickly filters and sorts simple comments.

XGBoost Calibrated;

Detects known risky or negative words.

RNN;

Understands full meaning, including sarcasm, emojis, and context.

Testing the models with real comments

The Test

Comparing all models against Messy live customer complaints.

The Discovery:

Standard Models: Missed technical crises like iPad app keeps closing because they lacked bad words.

The Sentinel: Correctly flagged the **intent**. It recognized a system crash as a brand risk.

The Strategic Edge:

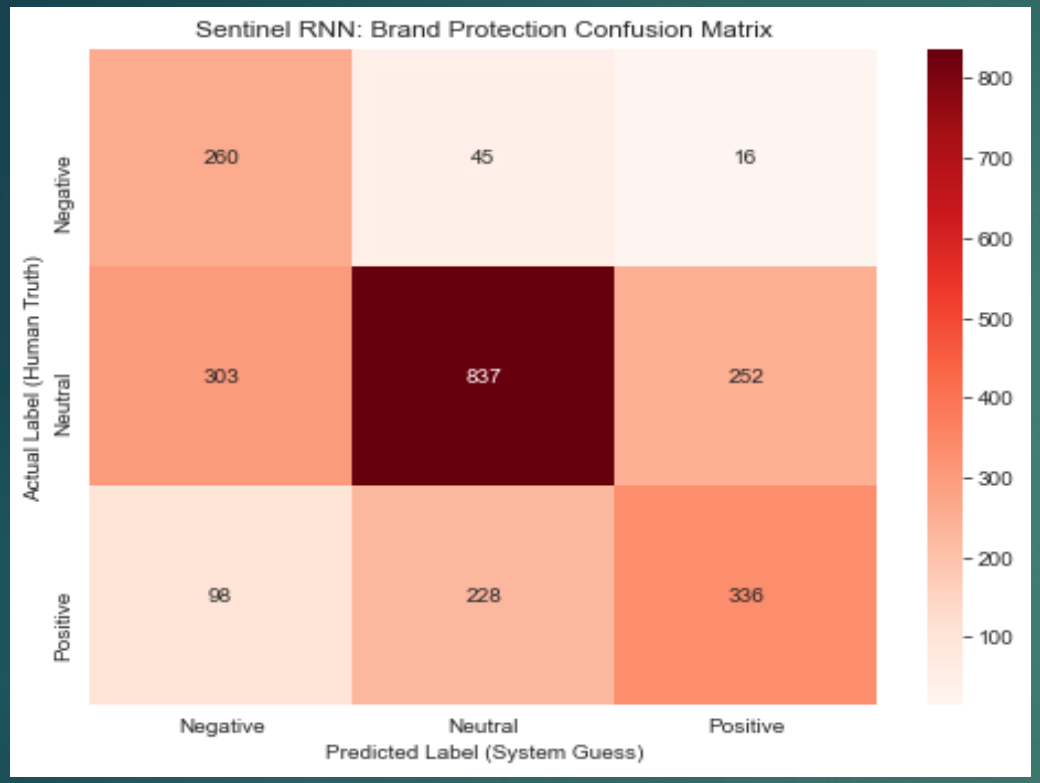
Beyond Keywords: Decodes sarcasm, slang, and technical frustration.

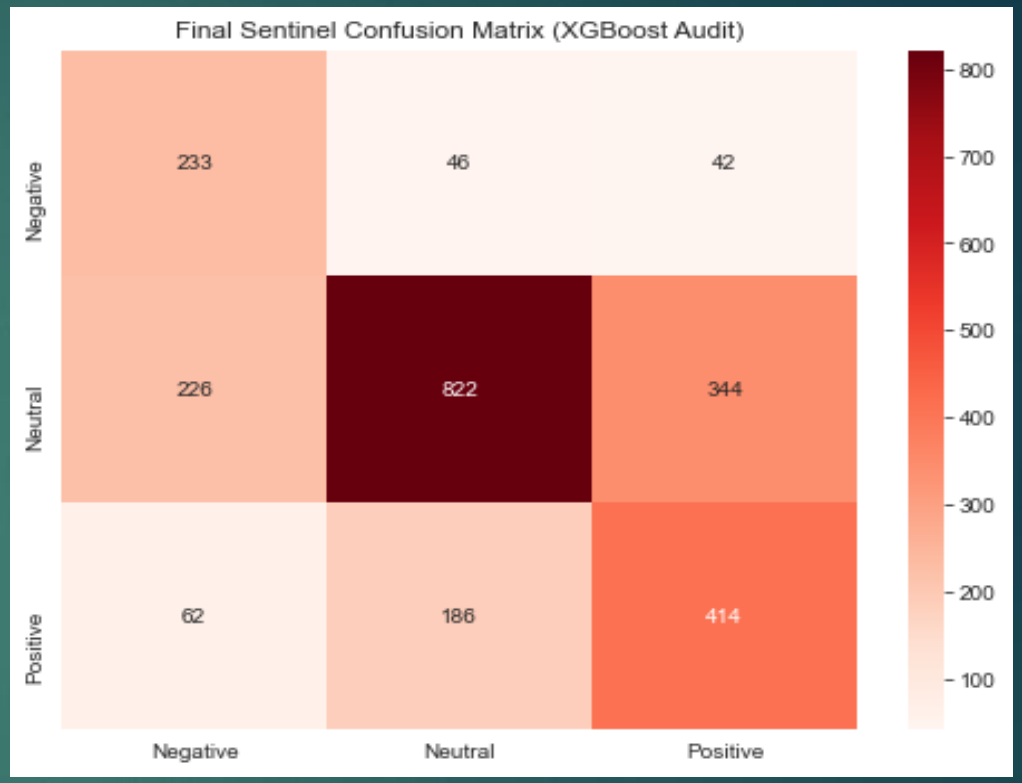
Risk Intelligence: Understands **what** is happening, not just **how** it is said.

```python
#real comments
test_comments = [
    "OMG my iphone is 🔥 but the battery is 💀... HELP!!",
    "RT @google: This new android app is awesome 💖 #SXSW",
    "I am SO mad at this apple iphone battery life! 👺",
    "Best app ever 💖 #SXSW",
    "Shiiiii I think I might switch to another phone.....this one is busted!!!!",
    "Oh great, another update that breaks my phone. Simply wonderful. 🙄", # Sarcasm Test
    "I think steve jobs is BEtter than mARK_2 Zuckenburg what you think",
    "The new UI is fine I guess, but it crashes twice a day. Very annoying."
]

# Execute comments
print("Below are the Verdicts made on the reckless comments by the 3 mdels")
for comment in test_comments:
    master_sentinel_test(comment)
```

```
Below are the Verdicts made on the reckless comments by the 3 mdels

COMMENT: "OMG my iphone is 🔥 but the battery is 💀... HELP!!"
MODELS          | VERDICT
-----------------------------------
Naive Bayes     | Negative
XGBoost         | Negative
Sentinel RNN    | Negative        (Risk Score: 58.4%)
🚩 ACTION: Brand Defense suggested by RNN.

COMMENT: "RT @google: This new android app is awesome 💖 #SXSW"
MODELS          | VERDICT
-----------------------------------
Naive Bayes     | Positive
XGBoost         | Positive
Sentinel RNN    | Positive        (Risk Score: 3.1%)
✅ STATUS: No immediate crisis detected.

COMMENT: "I am SO mad at this apple iphone battery life! 👺"
MODELS          | VERDICT
-----------------------------------
Naive Bayes     | Negative
XGBoost         | Negative
Sentinel RNN    | Negative        (Risk Score: 99.8%)
🚩 ACTION: Brand Defense suggested by RNN.

COMMENT: "Best app ever 💖 #SXSW"
MODELS          | VERDICT
```

Model Perfomance





After evaluating multiple models for sentiment analysis on tweets, the RNN is the most reliable Brand Sentinel. It balances ***catching harmful comments*** while still recognizing safe ones, making it the strongest model for brand protection.

It therefore becomes **the final decision-maker** because it understands language more deeply than the other models.

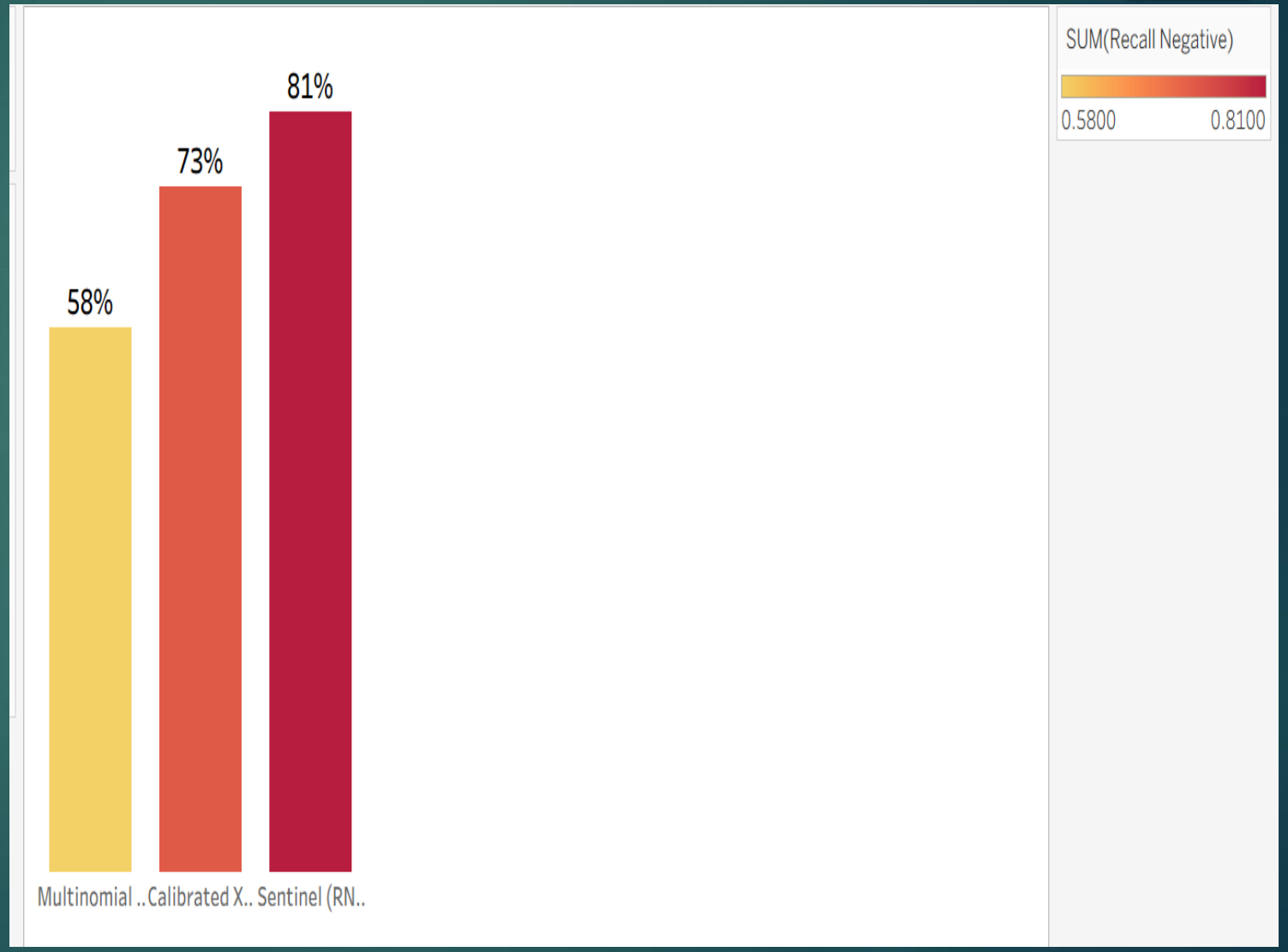
Why "Brand Sentinel" Works

Goal Met:

The system achieved **81% Negative Recall**, making it strong at detecting potential brand risks.

Safety First:

We prioritize finding negative comments over overall accuracy. In brand protection, missing a risk is worse than a false alarm.



Conclusion

24/7 Brand Equity Protection

Real-Time Speed: Detection in **seconds**, not days.

Superior Accuracy: **30% boost** in detection vs. standard models.

High-Alert Precision: **81% Recall** on critical brand threats.

Total Scalability: From 11,000 to **millions** of data points.

The Bottom Line:

The Brand Sentinel stays awake so the Brand managers can rest.

Thank You!